

VANTARA

Customer Intelligence Platform

FINAL PROJECT REPORT

Item	Details
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Project	End-to-End Customer Intelligence, Machine Learning and Analytics Platform
Core stack	Python 3.12, Pandas, scikit-learn, XGBoost, LightGBM, PyTorch, FastAPI, Streamlit, PostgreSQL, Docker
Final QA	49/49 tests passed; 77% src/ coverage; Ruff checks passed
Deployment	FastAPI, Streamlit and PostgreSQL verified with Docker Compose
Date	August 2026

Submission note: This report records the completed and verified project state. The CLV test R^2 limitation is documented transparently rather than reported as meeting the target.

1. Executive Summary

Vantara Customer Intelligence Platform is an end-to-end customer analytics system built on the Online Retail II dataset. It converts retail transactions into customer-level intelligence for churn prediction, Customer Lifetime Value (CLV) estimation, customer segmentation, anomaly analysis, explainability and business-facing analytics.

The completed solution includes a reproducible data pipeline, multiple classical machine-learning models, deep-learning experiments, K-Means and Gaussian Mixture Model clustering, SHAP/LIME/PDP explainability, a FastAPI service, PostgreSQL persistence, an interactive Streamlit dashboard, automated tests, code-quality checks and Docker Compose deployment.

Final verification item	Result
Single-command data pipeline	Passed - python run_pipeline.py
Automated tests	49 passed, 0 failed
src/ test coverage	77% (PRD target $\geq 70\%$)
Ruff linting	All checks passed
API p95 latency	7.38 ms locally (target < 400 ms)
Dashboard summary p95	6.46 ms locally (target < 3 s)
Docker Compose	API, dashboard and PostgreSQL verified running

2. Project Objectives and Scope

The project objective is to provide a practical customer-intelligence platform that supports retention and revenue decisions. The implemented scope covers the complete workflow from transaction-level data to business-facing predictions and analytics.

- Prepare and validate retail transaction data and engineer customer-level behavioral features.
- Predict churn and compare multiple classification approaches.
- Estimate CLV and identify high-value customers.
- Segment customers using unsupervised learning and provide business-readable profiles.
- Use ANN, LSTM and Autoencoder models for comparative deep-learning and anomaly analysis.
- Explain model behavior using SHAP, LIME and Partial Dependence Plots.
- Serve predictions through FastAPI and persist application records in PostgreSQL.
- Provide a Streamlit interface for prediction, explanation, segmentation, prioritization and analytics.
- Package the system for reproducible execution with configuration, tests, logging and Docker Compose.

3. Data Engineering and Feature Pipeline

The production data path begins with the two-sheet Online Retail II Excel file. day1_download.py combines the sheets, day2_clean.py applies the reusable cleaning logic, and day4_features.py generates the customer feature table. run_pipeline.py executes these stages in sequence through one command and emits structured INFO logs.

The pipeline produces a customer-level feature table of 4,996 customers. The project configuration file stores data paths, random seed, split proportions, churn-window settings, target metrics and feature configuration.

Feature group	Representative features
RFM / activity	Recency, Frequency, Monetary Total, Monetary Average
Basket / product	Average Order Value, Total Quantity, Average Basket Size, Unique Products
Temporal	Average Purchase Gap, Customer Lifespan Days
Behavioral	Return Rate, Price Sensitivity, Engagement Score
Value	Historical Revenue, Predicted CLV
Geography	Country
Target	Churn label based on the future purchase window

4. Modeling Approach and Results

4.1 Churn classification

Seven churn classifiers were consolidated in the final comparison: Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, SVM and ANN. The held-out test-set comparison selected the ANN as the recommended production churn model because it achieved the strongest overall balance of Accuracy, Recall, F1 and ROC-AUC.

Rank	Model	Accuracy	Precision	Recall	F1	ROC-AUC
1	ANN	0.7653	0.8045	0.8560	0.8295	0.8094
2	XGBoost	0.7507	0.7925	0.8480	0.8193	0.7982
3	Random Forest	0.7453	0.8294	0.7780	0.8029	0.8001
4	LightGBM	0.7267	0.8300	0.7420	0.7835	0.7974
5	Logistic Regression	0.7160	0.8409	0.7080	0.7687	0.7970
6	SVM	0.7080	0.8402	0.6940	0.7601	0.7945
7	Decision Tree	0.6720	0.7668	0.7300	0.7480	0.6453

4.2 Customer Lifetime Value regression

The CLV regression model achieved validation $R^2 = 0.6414$ and test $R^2 = 0.5883$. Test MAE was approximately 444.26 and RMSE approximately 770.98. The PRD target is $R^2 \geq 0.60$; the held-out test result is therefore slightly below target and is documented as a limitation. No artificial claim of target attainment is made.

4.3 Customer segmentation

The final clustering analysis compares K-Means and Gaussian Mixture Model (GMM) with seven clusters. K-Means provided substantially stronger internal separation (Silhouette 0.2954; Davies-Bouldin 1.0336) than GMM (Silhouette 0.0751; Davies-Bouldin 3.1545), so K-Means is the stronger clustering solution in the final analysis.

Segment	Customers	Avg Recency	Avg Monetary	Churn Rate	Business label
0	196	165.03	4,535.93	65%	High-Spend Occasional Customers
1	1,610	146.43	567.53	75%	Developing Customers
2	1,199	60.70	2,779.68	39%	Loyal Regular Customers
3	778	136.15	772.26	76%	Irregular At-Risk Customers
4	99	377.13	-843.75	98%	High-Return Lost Customers
5	223	25.88	22,904.41	8%	Champions
6	891	399.46	407.77	92%	Dormant Customers

4.4 Deep learning and anomaly analysis

The ANN achieved test ROC-AUC 0.8094 and F1 0.8295 and was selected as the final churn model. The LSTM purchase-behavior experiment achieved test Accuracy 0.7311, F1 0.8106 and ROC-AUC 0.7811. The Autoencoder anomaly workflow used a reconstruction-error threshold of 0.4082 and identified 250 of 4,996 customers (approximately 5.00%) as anomalous.

5. Explainable AI

Explainability is implemented using SHAP, LIME and Partial Dependence Plots. Saved artifacts include global SHAP importance, a SHAP summary plot, representative low-risk/borderline/high-risk customer explanations, a LIME local explanation, PDPs for influential variables and a plain-English explanation template.

The Streamlit Customer Explanation workflow now also supports lookup by Customer ID through the dashboard customer endpoint. Existing customer values can be loaded into the explanation workflow before generating churn explanations, satisfying the customer-search interaction required by the dashboard specification.

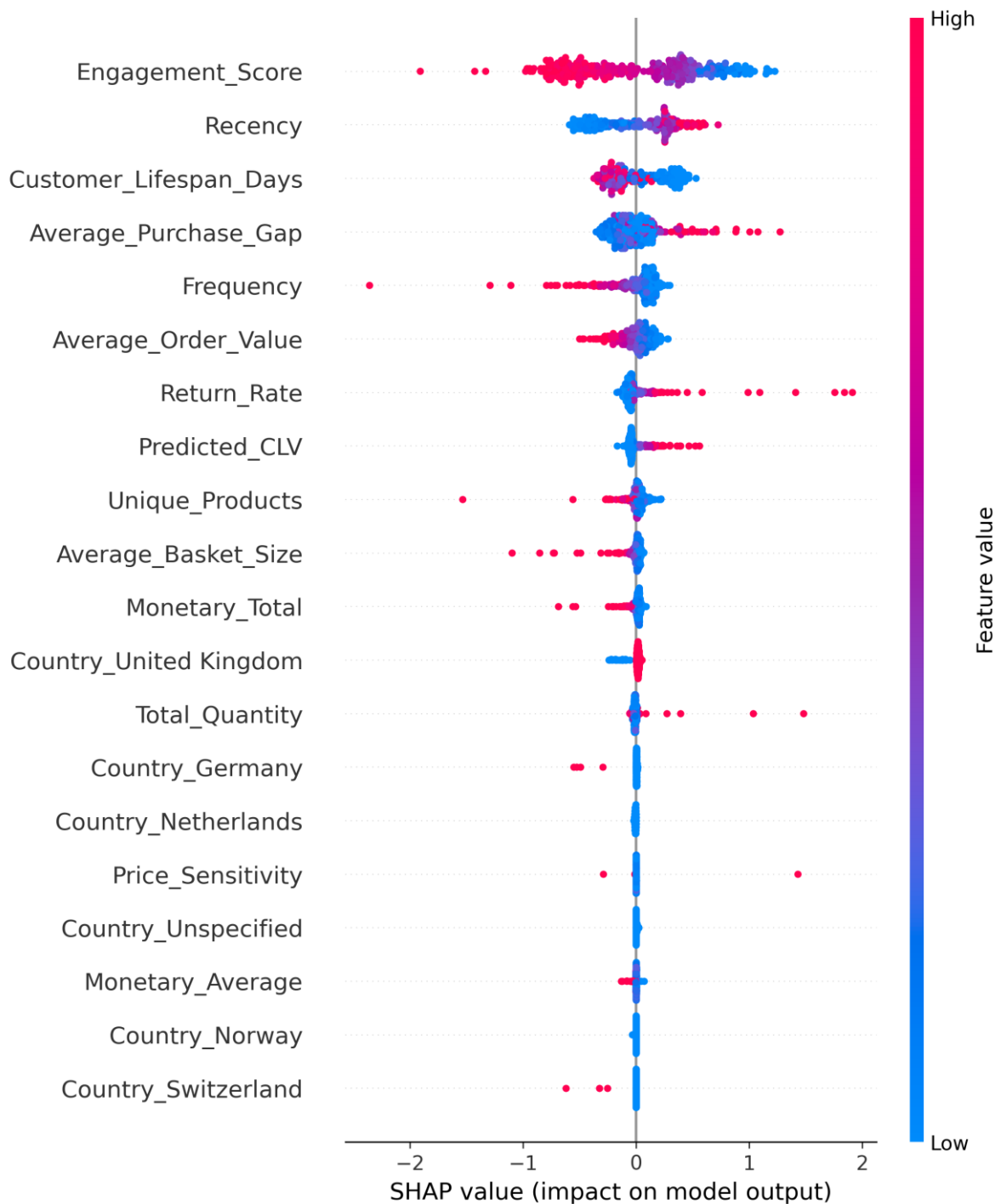


Figure 1. SHAP summary output generated by the project.

6. Application Layer: API, Database and Dashboard

6.1 FastAPI service

FastAPI exposes health, metadata, single prediction, SHAP-assisted explanation, batch CSV prediction, prediction history and dashboard analytics functionality. Swagger/OpenAPI documentation is available through the standard /docs route.

6.2 PostgreSQL persistence

SQLAlchemy models define Segment, Customer, Transaction and Prediction entities. PostgreSQL stores application data and prediction history. A database seed script loads customer profiles and business-readable segment information from the final segmentation output.

6.3 Streamlit dashboard

Dashboard view	Purpose
Executive Dashboard	KPIs, revenue trend, segment distribution and recent predictions
Churn Prediction	Single-customer churn scoring
Customer Explanation	Customer-ID lookup and feature-level churn explanation
Batch Prediction	CSV upload and downloadable prediction output
Prediction History	Stored prediction records
Customer Segmentation	Segment profiles and filters
Churn Leaderboard	Prioritization of high-risk/high-value customers
Revenue & CLV Analytics	Revenue trends, forecast and CLV analysis
System Status	Operational application/API status

7. Business Analytics

The final dashboard summary contains 4,996 customers, 3,329 high-risk customers, average CLV of approximately 27,325.98 and total revenue of approximately 20,972,968.14. Monthly revenue data spans December 2009 through December 2011. Combining churn probability, segment behavior and CLV supports retention prioritization toward customers who are both valuable and at risk.

8. Architecture and Workflow

The solution is organized into data, processing, modeling, serving, persistence and presentation layers. The repository contains the required system architecture, entity-relationship and workflow diagrams.

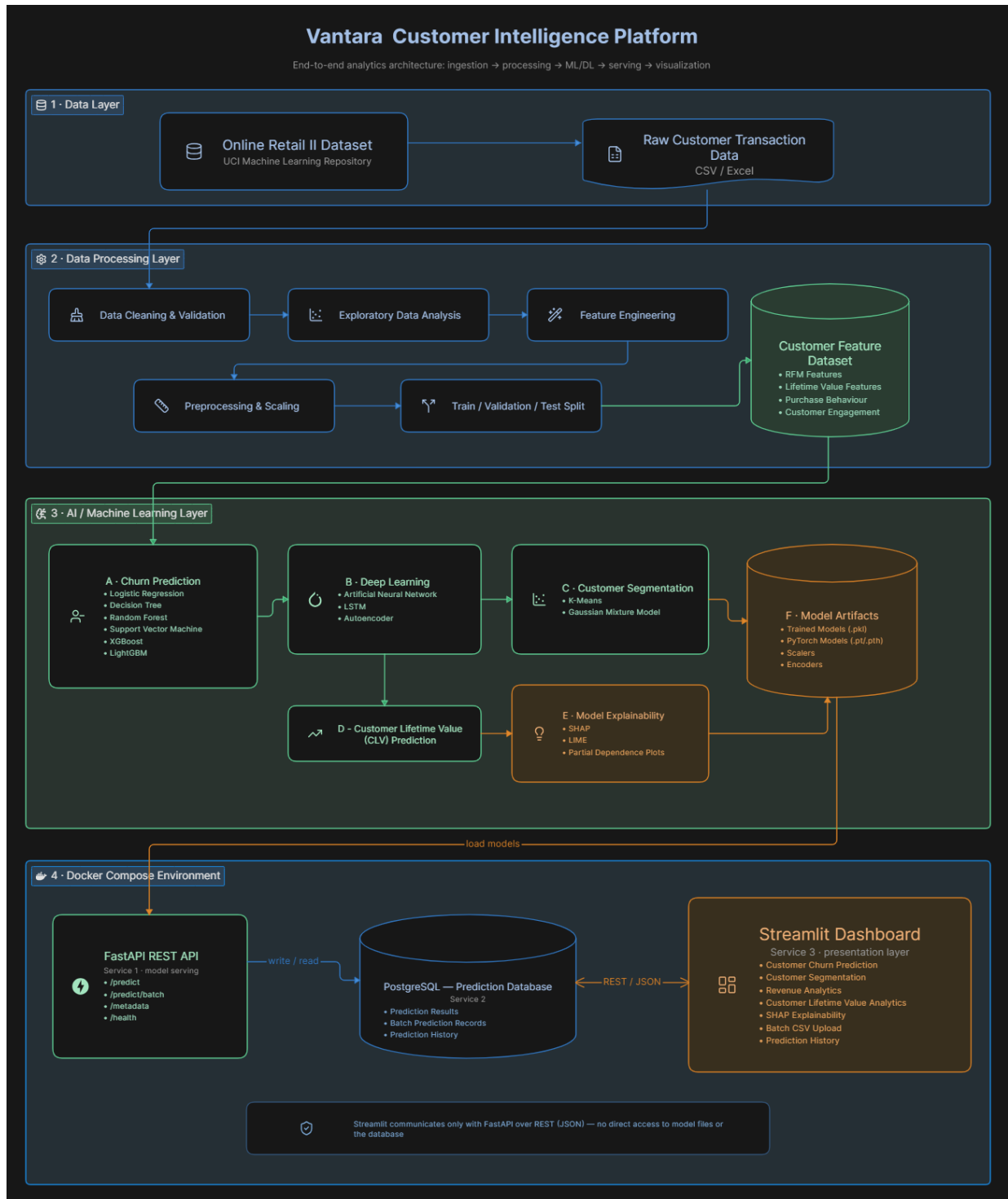


Figure 2. System architecture diagram.

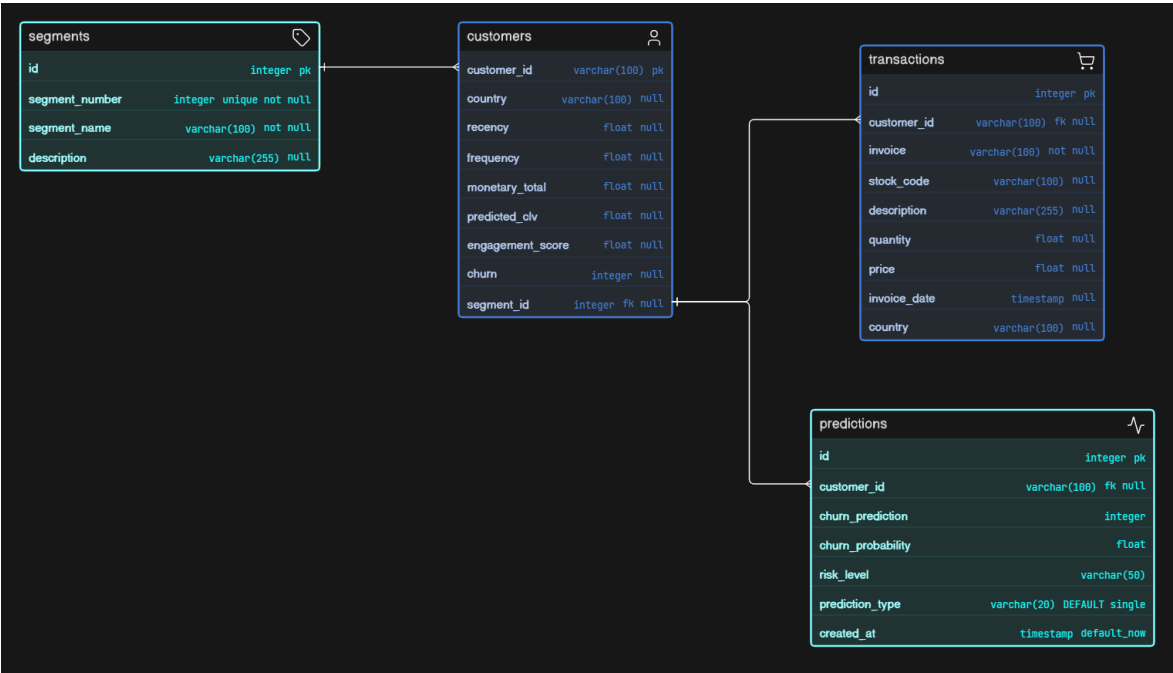


Figure 3. Entity-relationship diagram.



Figure 4. End-to-end project workflow.

9. Deployment and Reproducibility

The project can be executed locally with the Python environment or as a containerized stack. Docker Compose was verified to build and start the API, Streamlit dashboard and PostgreSQL services. The repository also includes requirements.txt, config/config.yaml, run_pipeline.py, structured pipeline logging and a README with local and Docker execution instructions.

Service	Role	Port	Final verification
vantara api	FastAPI backend	8000	Running
vantara dashboard	Streamlit UI	8501	Running
vantara postgres	PostgreSQL database	5432	Running

10. Testing, Quality and Performance Verification

The final pytest run collected 49 tests across API behavior, churn logic, data cleaning, feature engineering, extended feature engineering, pipeline helpers and preprocessing. All 49 tests passed. Coverage for src/ was 77%, exceeding the PRD threshold of 70%. Ruff was run after the final code changes and returned "All checks passed!".

Verification	Final result	Status
Tests	49 passed / 0 failed	PASS
src/ coverage	77%	PASS
Ruff	All checks passed	PASS
Single-command pipeline	Completed successfully	PASS
API p95	7.38 ms vs <400 ms target	PASS
Dashboard summary p95	6.46 ms vs <3 s target	PASS

The test run emitted third-party deprecation/future warnings. They did not cause test failures. Performance measurements are local verification measurements and should not be interpreted as production infrastructure benchmarks.

11. Limitations

Area	Final note
CLV regression	Test R^2 0.5883, slightly below the 0.60 target; documented rather than overstated.
Deep learning generalization	The available customer population is modest for deep-learning optimization; ANN/LSTM findings should be interpreted in that context.
Dependency warnings	The test suite passes, but third-party deprecation/future warnings remain.
Performance benchmark scope	Latency measurements were performed locally; cloud/production latency will depend on infrastructure and workload.

12. Future Enhancements

- Improve CLV modeling with additional behavioral features, alternative targets and longer customer history.
- Add automated model/data drift monitoring and scheduled retraining.
- Add CI/CD automation for tests, linting, image builds and deployment.
- Deploy the containerized application to a managed cloud environment.
- Add authentication/role-based access control for multi-user deployment.
- Expand sequence modeling when a larger longitudinal dataset becomes available.

13. Conclusion

Vantara Customer Intelligence Platform demonstrates a complete applied customer-analytics workflow from retail transactions to operational decision support. The final system combines reproducible data engineering, predictive modeling, CLV estimation, customer segmentation, explainability, API serving, database persistence, interactive analytics, automated testing, code-quality validation and containerized deployment.

The project meets the major engineering and analytical objectives while documenting the principal analytical limitation transparently: the held-out CLV R^2 of approximately 0.588 is close to but below the 0.60 target. With 49 tests passing, 77% source coverage, successful linting, successful single-command pipeline execution and verified Docker services, the project is in a completed submission-ready state.

Appendix A - Mathematical Foundations

A.1 Standardization

For feature x with mean μ and standard deviation σ : $z = (x - \mu) / \sigma$. Standardization prevents large-magnitude variables from dominating distance-based and gradient-based learning.

A.2 Logistic Regression

The churn probability is modeled by $p(y=1|x) = 1 / (1 + \exp(-(w^T x + b)))$.

A.3 Binary Cross-Entropy

For a binary target y and predicted probability p : $L = -[y \log(p) + (1-y) \log(1-p)]$.

A.4 Gradient Descent and Backpropagation

A generic parameter update is $w \leftarrow w - \eta \nabla L(w)$, where η is the learning rate. Backpropagation applies the chain rule to compute neural-network gradients.

A.5 Classification Metrics

Precision = $TP / (TP + FP)$; Recall = $TP / (TP + FN)$; $F1 = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall})$. ROC-AUC summarizes ranking/discrimination across classification thresholds.

A.6 Regression Metrics

MAE is the mean absolute prediction error; RMSE is the square root of mean squared error; $R^2 = 1 - SS_{\text{res}} / SS_{\text{tot}}$ measures explained variance relative to the mean baseline.

A.7 K-Means

K-Means minimizes within-cluster squared distance: $\sum_k \sum_{x \in C_k} \|x - \mu_k\|^2$.

A.8 Clustering Metrics

Silhouette score measures within-cluster cohesion relative to neighboring-cluster separation (higher is better). Davies-Bouldin compares within-cluster scatter with between-cluster separation (lower is better).

A.9 SHAP

SHAP assigns additive feature contributions based on Shapley values from cooperative game theory, allowing a prediction to be expressed as a baseline plus feature contributions.

A.10 Autoencoder Reconstruction Error

An autoencoder learns to reconstruct normal inputs. Reconstruction error can be expressed as mean squared error between input x and reconstruction $\hat{x}$; unusually high error is used as an anomaly signal.

Appendix B - Final Deliverables Checklist

Deliverable	Final status
Cleaned, feature-engineered dataset and documented pipeline	Completed
Single-command raw-to-feature pipeline	Completed and verified
Classical churn-model comparison	Completed
CLV regression	Completed; test R^2 limitation documented
K-Means + second clustering method	Completed (K-Means + GMM)
Business-readable segmentation profiles	Completed
ANN, LSTM and Autoencoder	Completed
SHAP, LIME and PDP explainability	Completed
Customer-ID explanation workflow	Completed

FastAPI service / Swagger	Completed
PostgreSQL persistence	Completed
Streamlit dashboard / batch download	Completed
Docker Compose deployment	Completed and verified
Architecture / ER / workflow diagrams	Completed
Automated tests	49/49 passed
Source coverage	77% - target achieved
Ruff code-quality check	All checks passed
README / reproducibility instructions	Completed
Final written report	Completed (this document)